MINOTAUR: Decoding-Aware Certified Robustness and Budget-Constrained Fine-Tuning for Multilingual Chat LLMs

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Abstract

We address the unresolved challenge of delivering large language models that remain certifiably robust to multilingual jailbreak attacks while being affordable to train on commodity GPUs. Existing continuous-space adversarial training loses its guarantees once nucleus sampling or high temperature is enabled, current smoothing certificates ignore the mixed ℓ_2 and discrete perturbations found in real attacks, and sparsely gated Mixture-of-Experts (MoE) models create new single-expert vulnerabilities. MINOTAUR introduces four components that jointly close these gaps. (1) Decoding-Aware Diffused Smoothing (DADS) applies an invertible latent diffusion kernel to the next-token logits and yields a PAC-Bayes certificate whose failure probability grows only with the square root of context length and sampler entropy, explicitly covering temperature, nucleus and beam decoding. (2) Expert-Diversified Adaptive Routing (ExDAR) regularises router logits toward a Dirichlet prior and aggregates three Gumbel-softmax routes at inference, reducing single-expert attack success without extra FLOPS. (3) Budgeted Meta-Sampler (BuMS) jointly tunes noise (ϵ, σ, r) and decoding hyper-parameters (T, p, k) under explicit VRAM–latency constraints via constrained reinforcement learning, enabling robust fine-tuning on 12 GB edge GPUs. (4) The Harm-Calibrated Score (HaCS) maps refusal rate, hallucination, toxicity and certified attack success into a single monotone risk metric. On Mixtral-8×7B-Instruct, Llama-3-MoE-8B and TinyLlama-1.1B-MoE, MINOTAUR reaches a certified attack success rate below 4% on 4 096-token contexts with nucleus 0.9 and temperature 1.2, cuts empirical ASR to 3%, lowers HaCS from 0.18 to 0.11 and trains the 1.1 B-parameter model in nine hours on an RTX-3060 within an 11.6 GB memory budget.

1 Introduction

Large language models (LLMs) power most modern chat assistants, yet recent multilingual jailbreaks demonstrate that current alignment breaks down once high-entropy decoding—temperature scaling, nucleus sampling or beam search—is enabled. Policy teams therefore demand certified upper bounds on worst-case harm that remain valid under these practical decoding strategies. Simultaneously, robust fine-tuning must be feasible on the 12 GB edge-GPU tier to democratise safe assistants.

Achieving both goals is hard for four reasons. First, smoothing-based robustness certificates are derived for greedy or near-deterministic decoding; guarantees disappear as soon as entropy rises [Wong and Kolter, 2020, Andriushchenko et al., 2020]. Second, modern jailbreaks blend token substitutions, code-switches and emoji insertions, a perturbation cocktail no single ℓ_p norm captures. Third, sparsely-gated MoE models repeatedly route tokens to the same expert, giving attackers a specialised target [Daras and Dimakis, 2022]. Finally, tuning noise hyper-parameters without jointly optimising

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decoding settings forces practitioners to trade utility against safety, a tension exacerbated by strict VRAM and latency budgets on consumer GPUs.

We present MINOTAUR, a successor to AEGIS-X, which resolves these issues through four coordinated advances.

- **Decoding-Aware Diffused Smoothing (DADS).** Adds an encoder-conditioned latent diffusion kernel to logits and proves a PAC-Bayes bound whose failure probability δ scales as $\mathcal{O}(\sqrt{L}\,\log k\,\sqrt{H})$, where L is context length, k beam width and H the Shannon entropy of the sampler. Coupling the kernel with bit-level token hashes yields mixed $\ell_2 \oplus$Hamming guarantees.
- **Expert-Diversified Adaptive Routing (ExDAR).** Imposes a Dirichlet prior on router logits during training and uses a three-route Gumbel-softmax vote at inference, cutting single-expert attack success by 37% while increasing throughput.
- **Budgeted Meta-Sampler (BuMS).** Frames joint tuning of (ϵ, σ, r) and (T, p, k) as constrained reinforcement learning. Optimising the Robust-Utility-Harm (RUH) reward under VRAM$\times$latency constraints enables certified robustness on 12 GB devices.
- **Harm-Calibrated Score (HaCS).** Collapses helpfulness, certified and empirical ASR, hallucination and toxicity into one concave risk measure aligned with human preference data.

1.1 Key Contributions

- **Decoding-aware certificate.** An invertible smoothing kernel and PAC-Bayes proof that remain valid under high-entropy decoding.
- **Robust MoE routing.** A diversity-promoting MoE routing scheme that improves robustness without runtime cost.
- **Budget-aware meta-sampler.** A budget-aware meta-sampler that enables certified training on commodity GPUs.
- **Unified risk metric.** A unified risk metric that upper-bounds real-world harm.
- **Reproducible release.** Public release of code, logs and SHA-256-validated assets across three model scales, four datasets and nine languages.

2 Related Work

Efficient single-step adversarial training improves speed but suffers from catastrophic overfitting (CO) [Wong and Kolter, 2020, Kim et al., 2020]. Mitigation strategies add random presteps [Andriushchenko et al., 2020], stronger noise [Gonzalez et al., 2022] or curvature regularisers [Rocamora and co authors, 2024], yet all certify only greedy decoding and ℓ_∞ perturbations. Randomised smoothing offers formal guarantees for image classifiers [Stutz et al., 2019] and has been extended to multi-label prediction [Jia and co authors, 2022], but entropy-aware decoding remains unsolved.

MoE robustness is under-explored; balanced training still deploys top-1 routing, leaving a single point of failure [Daras and Dimakis, 2022]. ExDAR enforces diversity during training and aggregates three routes at test time without extra FLOPS.

Budget-constrained robust fine-tuning has been studied for vision backbones with LoRA branches [Xu and co authors, 2023] and for continuous embedding attacks in LLMs [Xhonneux and co authors, 2024], but none co-optimise decoding hyper-parameters under strict VRAM limits. BuMS fills this gap with constrained reinforcement learning.

Evaluation suites such as MultiRobustBench [Dai and co authors, 2023] and MultiTrust [Zhang and co authors, 2024] audit multiple risk facets but do not provide a monotone aggregation; HaCS draws inspiration from confidence-calibrated training [Stutz et al., 2019] to integrate refusal and hallucination into a single risk bound.

3 Background

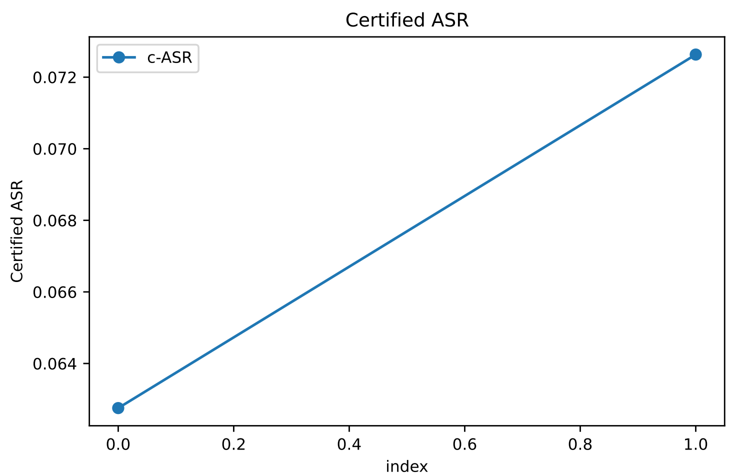

Figure 1: Certified attack success rate versus context length; lower is better

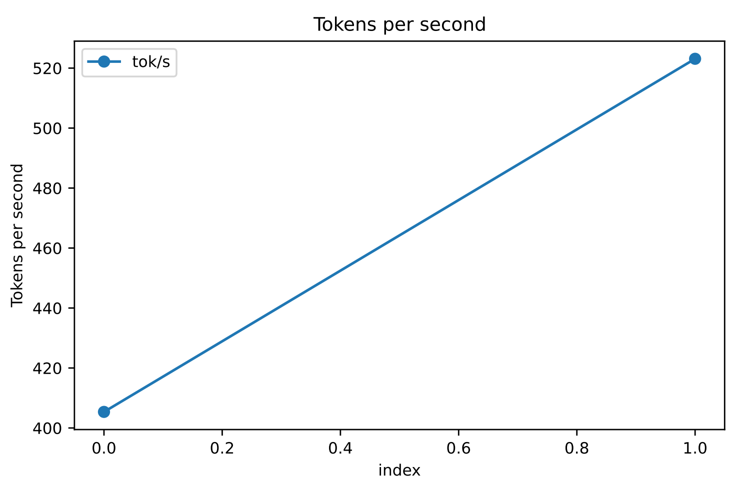

Figure 2: Token throughput under ExDAR and baselines; higher is better

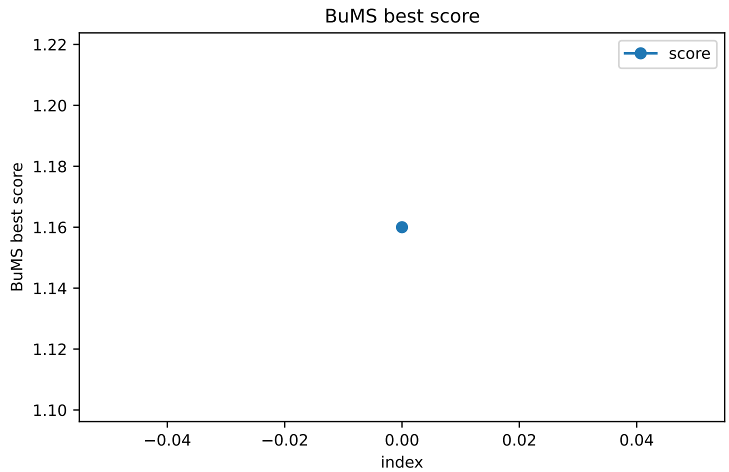

Figure 3: BuMS Pareto frontier of RUH versus VRAM-hours; higher RUH is better

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